

Samanvay Malapally Sudhakara

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EDUCATION

MS Financial Engineering (GPA 3.77/4.00), <i>University of Illinois Urbana-Champaign, Champaign, IL</i>	May 2024
BE Mechanical Engineering , <i>Ramaiah Institute of Technology, Bangalore, India</i>	July 2021

TECHNICAL SKILLS

Languages	Python, C++, Rust, SQL, R, Linux Shell
Data & Visualization	NumPy, pandas, Seaborn, Matplotlib, Plotly, ggplot2, Tableau, Power BI, Streamlit, Excel
Machine Learning	Regression, Classification, Tree-Based Ensemble Models, Reinforcement Learning (MAB, DQN, DDPG, PPO, SARSA)
Generative AI	LLM Training, Inference (vLLM, llama.cpp, SGLang, Ollama), Fine-Tuning (LoRA, QLoRA, GRPO), Quantization (GPTQ, AWQ), Speculative Decoding (MTP, DFlash), MCP Construction, Evals & Multi-Agent Orchestration
ML Packages	PyTorch, TensorFlow, JAX, scikit-learn, Transformers, Hugging Face Hub
Gen AI Packages	vLLM, MCP, Pydantic AI, LlamaIndex, Qdrant, Weaviate, RAG/GraphRAG, Phoenix/OpenInference, Claude Code, Codex
Cloud & Databases	GCP (Compute Engine, GCS, Vertex AI, Cloud Run, BigQuery, ADK), AWS (S3, EC2, Redshift, SageMaker, Lambda, Strands)
Data Engineering & CI/CD	Git, GitHub, GitLab, Docker, Snowflake, dbt, MySQL, Apache (Hadoop, Airflow, Spark, Kafka)
Quantitative Skills	Stochastic Calculus, Derivative Pricing, Hypothesis Testing, Statistical Analysis, Risk Management, Market Microstructure, Algorithmic Trading, Fixed Income & Interest Rate Models, Game Theory & Auction Theory

PROFESSIONAL EXPERIENCE

Quantitative Associate , <i>Northwestern Mutual Investment Management Company LLC, Milwaukee, WI</i>	Oct 2024 – Present
- Quant on the Public Fixed Income team (\$140B+ AUM) spanning the full fixed income spectrum: IG/HY corporates, EM sovereign & corporate credit, MBS, CMBS, CLOs, ABS, munis, rates & macro, and fixed income derivatives, plus the affiliated CLO manager, 720 East (~\$4.2B AUM). - Consolidated analytics data from BlackRock Aladdin Explore, Bloomberg, eMBS, Virtus CLO & internal sources into unified Snowflake tables & views modeled with dbt, backed by a comprehensive data test suite & data-integrity monitors. - Led the department's shift from Tableau, Power BI & Excel reporting to a multi-app Streamlit interface for dashboards & analytics, with save-and-load for individual charts, enabling faster, more programmatic & interactive decision making. - Designed a semantic-layer-based query engine behind the Streamlit apps, where column configs drive display, aggregation & lookback rules and Jinja conditionally assembles the SQL, giving users on-the-fly control over grouping, filtering & analytics like Brinson attribution & Frongello linking at up to 5x the performance of a Snowpark approach. - Built a connected suite of tools for portfolio managers, analysts & traders that collapses the ideation-to-trade workflow from hours to minutes. - Built Agency MBS analytics (CPR & S-curve analysis, WALA ramps) over custom cohorts built from loan-level data, benchmarking a CUSIP's prepayment behavior against its broader cohort. - Built factor-based models grounded in Barclays QPS research, drawing on factors such as EMC, ESP & SPIDER, to generate systematic buy & sell signals and give PMs a quantitative overlay for security selection & relative-value decisions. - Automated prior-day analytics ingestion with Selenium & ChromeDriver, cutting daily ingestion time by 4 hours & delivering validated data to these tools before market open. - As part of the AI working group, drove broader AI adoption across the floor, helping users automate existing workflows with enterprise AI tools like Copilot Studio & Bloomberg's ASKB.	
Quantitative Analyst , <i>JIA Finance, New York, NY</i>	Jan 2023 – Aug 2023
- Automated mortgage guideline interpretation using the OpenAI GPT-3.5 & GPT-4 APIs with LangChain & FAISS/Chroma vector stores for rapid ingestion & validation of documents, improving the speed & accuracy of mortgage processing. - Engineered specialized, task-specific Retrieval-Augmented Generation (RAG) Q&A chains that refined query processing through a recursive tree approach combined with LLM-based memoization, boosting zero-shot accuracy by 30%. - Ran exploratory data analysis on over 100M Fannie Mae mortgage records in Tableau, with data cleaning & feature engineering via AWS SageMaker & the Redshift connector, to surface actionable insights. - Modeled loan survival & default rate curves using the Cox proportional hazards model & the Kaplan–Meier estimator, & assessed the impact of macroeconomic factors such as unemployment & income on default & delinquency rates. - Crafted a comprehensive cash flow model for mortgage-backed securities (MBS), integrating credit default, prepayment & loss severity models with a yield-curve-calibrated Cox–Ingersoll–Ross (CIR) interest rate model.	

PROJECTS

High Frequency Trading Backtesting Pipeline with RL Trading Bot
- Built parsers for tick, depth-update & market-by-order data from sources such as CME Globex, IEX, Binance & OANDA, with cleaning & exploratory analysis to identify outliers, spot missing data & visualize the data. - Simplified strategy backtesting with command-line pipelines driven by Linux shell scripts that integrate the C++ & Python components needed to run backtests, process results & tune hyperparameters. - Engineered DQN & DDPG models in C++ for a market-taking agent & a market-making agent, using the PyTorch C++ API for the models & SQLite for the experience replay store, backtested on the pipelines above.

Forex Ladder Trading Strategy (*practicum sponsored by BP Trading*)

- Enhanced return potential by developing a forex ladder trading strategy using quantitative techniques for grid & lot-sizing optimization.
- Sped up backtests by up to 2000x using just-in-time compilation on tick data across major currency pairs.
- Tuned proprietary trading models with automated hyperparameter search (Optuna, PySwarms, DEAP genetic algorithms) & validated them using walk-forward analysis & Monte Carlo simulation.

Agentic Equity Research Pipeline on Self-Hosted LLM Inference

- Deployed a 35B-parameter MoE model (Qwen3.5-A3B) on an NVIDIA DGX Spark (GB10/ARM64, 128GB unified memory) via vLLM with NVFP4 quantization & DFlash speculative decoding, fronted by a Pydantic AI + MCP agent gateway (streamable HTTP) exposed through a Cloudflare Tunnel with Access JWT for zero-open-port ingress & instrumented with Phoenix/OpenInference.
- Engineered an OCR pipeline with built-in verification over a BSE/NSE filing sentinel, using PaddleOCR & LlamaParse (LlamaIndex) for document processing & the self-hosted Qwen model for agentic verification of extracted figures; cleansed data lands in BigQuery tables with a full observability & verification trace, exposed over REST API & MCP.
- Architected an agentic equity analyst with human user-bias conditioning that analyzes tickers using the self-hosted LLM, grounded on the cleansed financial data plus a Qdrant RAG engine & web search for the latest news, with Phoenix observability across the run, to generate research reports & recommendations.

CERTIFICATIONS

- **CFA level 3 Candidate** — *CFA Institute*
- **Bloomberg Market Concepts** — *Bloomberg for Education*
- **Machine Learning Specialization** — *DeepLearning.AI*
- **Deep Learning Specialization** — *DeepLearning.AI*
- **Retrieval Augmented Generation** — *DeepLearning.AI*